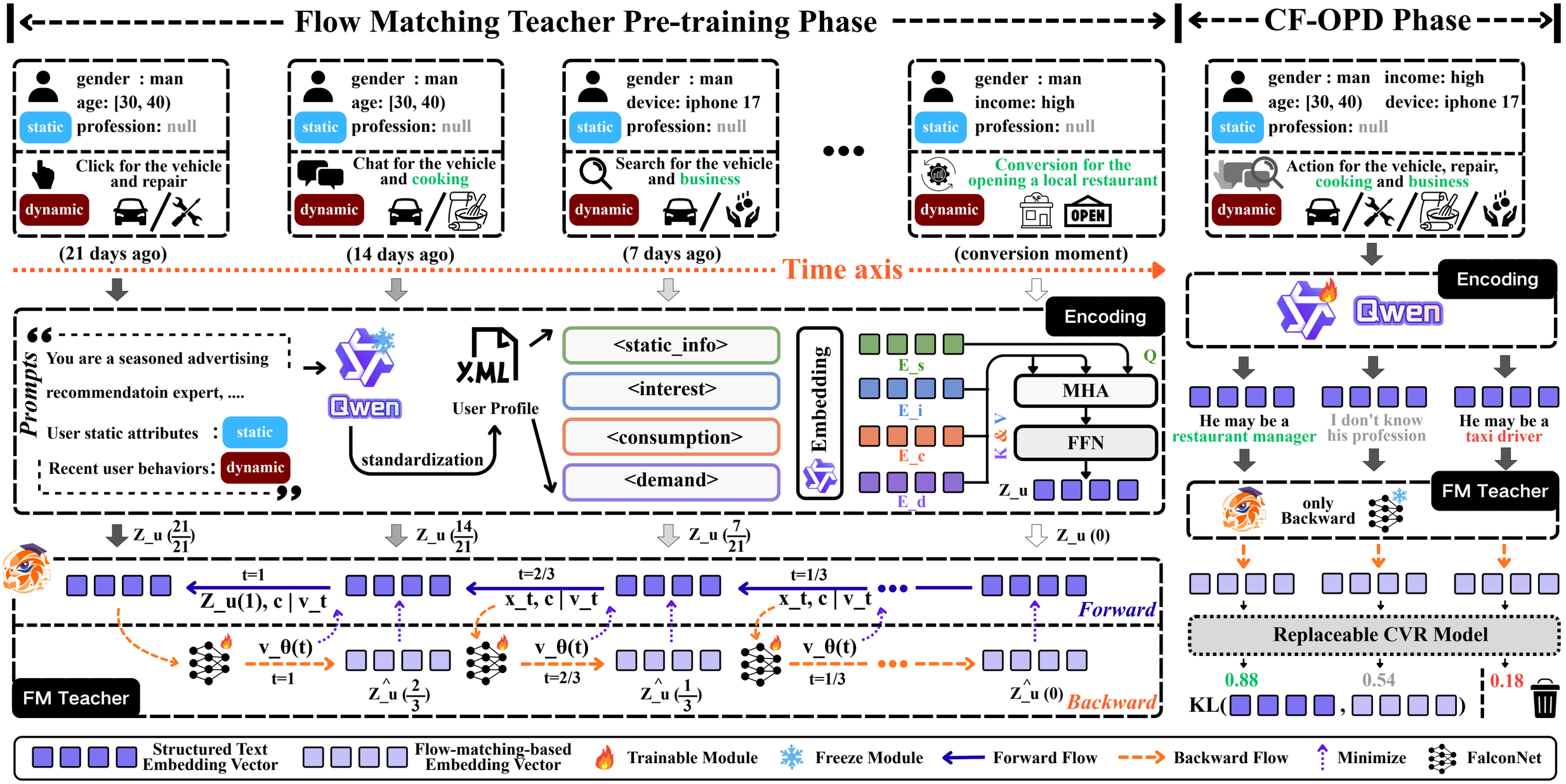

Flow Matching Teacher Pre-training Phase
CF-OPD Phase
gender : man
age: [30, 40)
static
profession: null
Click for the vehicle and repair
dynamic
(21 days ago)
gender : man
age: [30, 40)
static
profession: null
Chat for the vehicle and cooking
dynamic
(14 days ago)
gender : man
device: iphone 17
static
profession: null
Search for the vehicle and business
dynamic
(7 days ago)
gender : man
income: high
static
profession: null
Conversion for the opening a local restaurant
dynamic
OPEN
(conversion moment)
Time axis
gender : man
income: high
age: [30, 40)
device: iphone 17
static
profession: null
Action for the vehicle, repair, cooking and business
dynamic
Encoding
Prompts
You are a seasoned advertising recommendatoin expert, ...
User static attributes : static
Recent user behaviors: dynamic
Qwen
standardization
XML
User Profile
<static_info>
<interest>
<consumption>
<demand>
Embedding
E_s
E_i
E_c
E_d
Q
K&V
MHA
FFN
Z_u
Encoding
Qwen
He may be a restaurant manager
I don't know his profession
He may be a taxi driver
FM Teacher
only Backward
Z_u (21/21)
Z_u (14/21)
Z_u (7/21)
Z_u (0)
t=1
Z_u(1), c | v_t
t=2/3
x_t, c | v_t
t=1/3
x_t, c | v_t
Forward
v_θ(t)
t=1
v_θ(t)
t=2/3
v_θ(t)
t=1/3
Z^_u (2/3)
Z^_u (1/3)
Z^_u (0)
Backward
FM Teacher
Replaceable CVR Model
0.88
0.54
0.18
KL(,)
Structured Text Embedding Vector
Flow-matching-based Embedding Vector
Trainable Module
Freeze Module
Forward Flow
Backward Flow
Minimize
FalconNet